STRATEGY RECOMMENDATION · WORKPLACE AI

When better AI stops paying

THE QUESTION

When will AI hit diminishing returns at work, and when should companies redesign the work instead?

OUR ANSWER

Around 4x: by then ~60% of AI's possible gain is in; checking and coordination cap the rest. In software and finance, redesign already saves ~1.7x the hours of a model upgrade.

INSIDE

01 Why now · time map

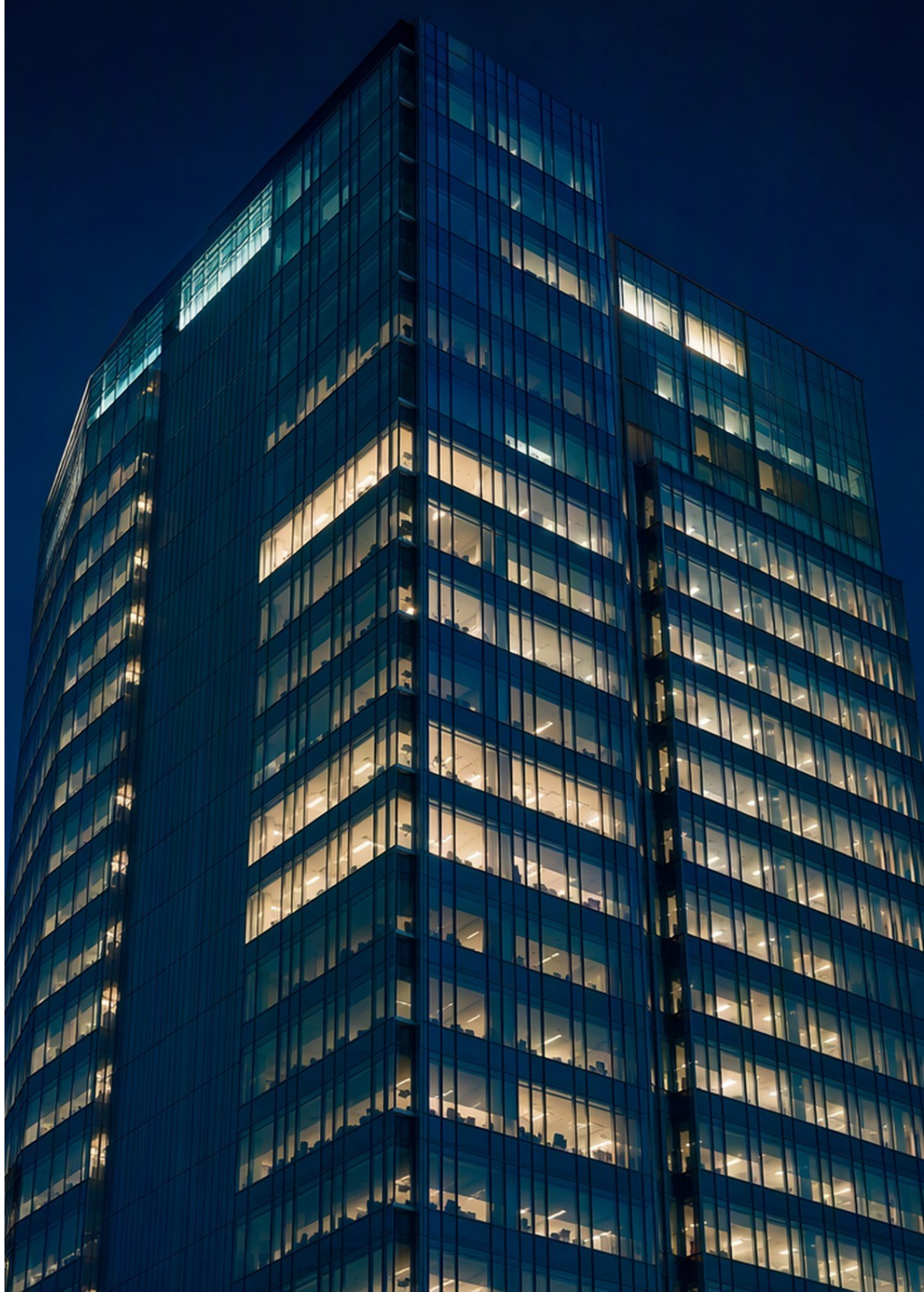
02 Curve · four forces

03 Better AI or redesign

04 Pilot, risks, evidence

Evan Chan · October 2026

Built from public information only · Model and estimates are the author's own



EXECUTIVE SUMMARY

Redesign the work around AI before buying the next model: it saves ~1.7x the hours in software and finance

1 The problem

AI speeds up an estimated 35–45% of the hours. Checking and hand-offs don't, so AI alone caps output gains at +39–62%.

2 Why now

AI spending heads to $2.7T in 2026 (Gartner), yet 9 in 10 executives see no productivity impact so far (NBER).

3 What we recommend

Run a next-dollar test per workflow. Fund redesign first where checking + coordination exceed ~1.5x AI-assisted time.

4 How we'll know

12-week pilots vs. a same-AI control: ≥12% fewer hours, ≥30% less checking, ≥60% of freed hours put to named use.

BOTTOM LINE

Better AI flattens at about 4x; redesign is now the bigger lever.
By 4x, ~60% of AI's possible gain is in. Support still gains from its next upgrade.

THE ASK

Two weeks of time sampling, then three 12-week redesign pilots. Hold upgrades outside support until the test runs.

EXPECTED IMPACT

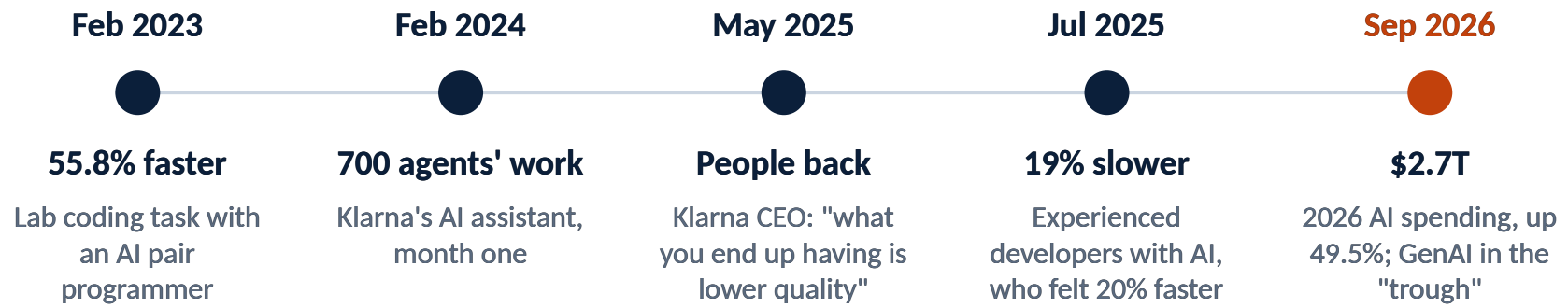
13–19% fewer hours per unit on top of today's AI savings (author's model).

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THE PROBLEM · WHY NOW

AI spending is heading to $2.7 trillion in 2026, yet 9 in 10 executives report no productivity impact so far

HOW THE GAP OPENED



WHY IT MATTERS NOW

9 in 10

executives at ~6,000 firms saw no AI effect on productivity or jobs over 3 years

NBER WP 34836, Feb 2026

~40%

of the time AI saves is lost to rework: checking, correcting, rewriting

Workday, 3,200 staff, Jan 2026

2.8%

of work hours saved by chatbot users, with no measurable effect on earnings or hours

NBER WP 33777, 25,000 workers

BOTTOM LINE

Task gains are real; firm-level gains aren't showing up yet.

The gap sits in the work around the AI: checking, hand-offs and approvals, which a model upgrade doesn't touch.

WHAT'S COMING NEXT

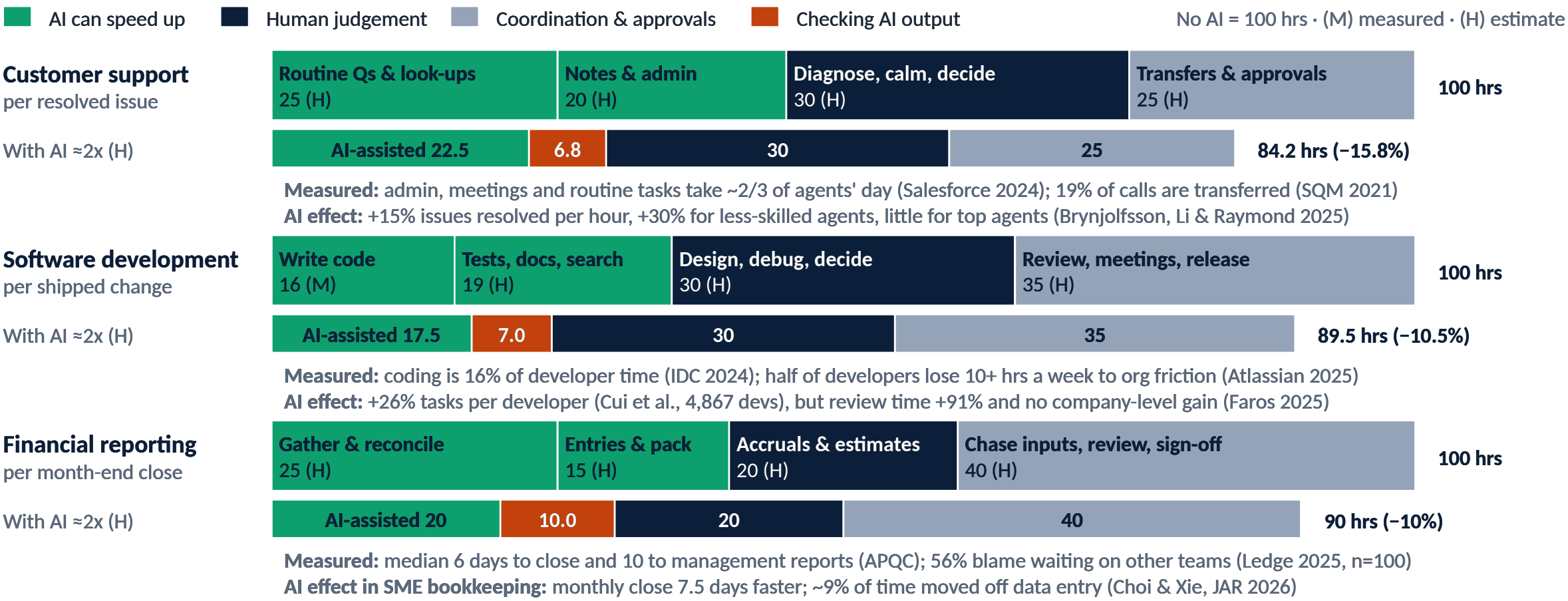
Agents. Gartner expects AI to resolve 80% of common service issues by 2029; Google says 75% of its new code is AI-generated. More output means more to check and coordinate.

Sources: Peng et al. (Feb 2023); Klarna (Feb 27, 2024); Bloomberg via CX Dive (May 2025); METR (Jul 10, 2025); Gartner (Sep 16, 2026; Mar 5, 2025); NBER WP 34836 (Feb 2026); Workday (Jan 14, 2026); NBER WP 33777 (rev. Mar 2026); Business Insider (Apr 22, 2026).

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THE PROBLEM · WHERE THE TIME GOES

By our estimate, AI can speed up only 35–45% of the hours; at today's ≈2x, total time falls just 10–16%



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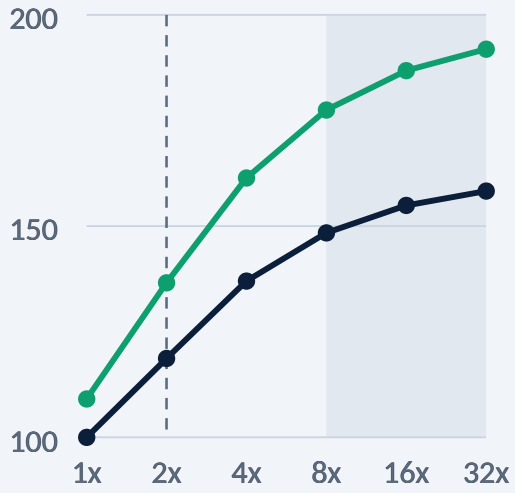
DIAGNOSIS · OUTPUT AS AI IMPROVES

Better AI flattens fast: by 4x it delivers ~60% of its possible gain, and alone it caps output at +39–62%

Better AI only · Better AI + redesign · Today ≈2x (H) · Each doubling adds under 5%

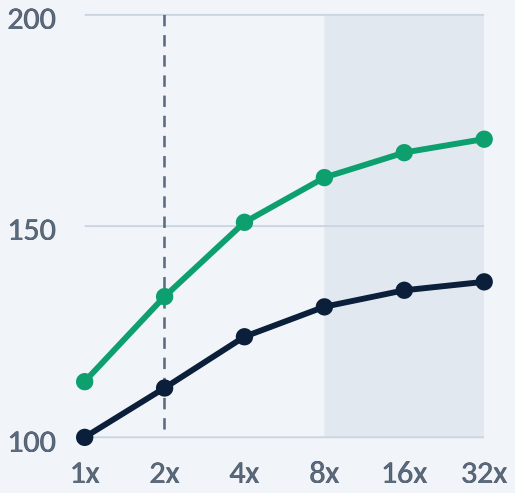
Customer support

AI alone: ceiling +62%
With redesign: +97%



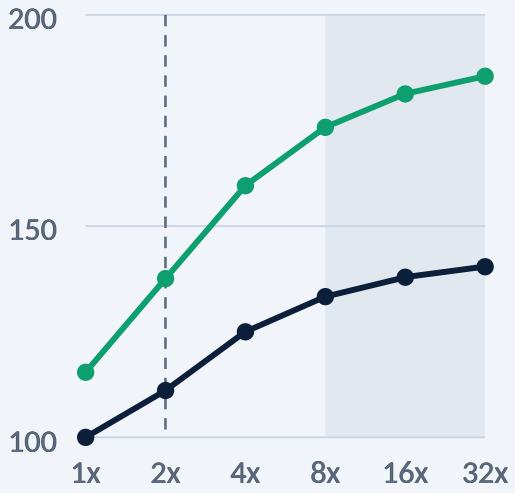
Software development

AI alone: ceiling +39%
With redesign: +74%



Financial reporting

AI alone: ceiling +43%
With redesign: +90%



Output per hour, no AI = 100. Author's model: hours = AI-assisted work ÷ speed-up + checking + judgement + coordination (splits on slide 4). Redesign = ⅓ less coordination, 40% less checking (H).

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WHERE IT BREAKS

By 4x, about 60% of AI's possible gain is in.

The next doubling adds 6–8%; each one after, under 5%. Redesign lifts the ceiling itself: +74–97% vs. +39–62% for AI alone.

THE DATA AGREE

Team output already looks flat: AI agents lifted pull requests 23% but not resolved issues (718 firms, Chen & Stratton 2026). Executives expect just +1.4% productivity in 3 years (NBER).

ROOT CAUSES · FOUR FORCES

Four forces cap the gain, and checking alone eats about 40% of the time AI saves

1 The automation ceiling

AI can speed up only 35–45% of hours.

Coding is 16% of developer time (IDC). Only 14% of service issues are fully resolved in self-service (Gartner). In the close, 56% cite waiting on other teams (Ledge).

Observed · splits on slide 4 are (H)

2 The verification burden

Checking eats ~40% of the time saved.

Nearly 4 in 10 hours AI saves go to rework (Workday, 3,200 staff). Developers spent 9% of time reviewing AI code (METR); CFOs, 26% of their day checking AI output (Datarails).

Observed · Datarails is vendor data, directional

3 The organizational bottleneck

Queues absorb the speed-up.

As pull requests merged rose 98%, review time rose 91%, with no company-level gain (Faros, 10,000+ devs). AI-written PRs wait 4.6x longer for review (LinearB).

Observed · vendor telemetry, directional

4 The moving target

Saved time goes to more work, not less.

85% of Danish AI users moved saved time to other tasks; under 10% took breaks or leisure. At one tech firm, AI raised pace, scope and hours (HBR 2026).

Observed · (H) whether the extra work is higher-value

WHAT WE'D NEED TO CONFIRM

Each (H) split on slide 4, from two weeks of time sampling per workflow · checking minutes per AI output, by risk tier · review, approval and input wait times from ticket, Git and close-calendar logs · where freed hours went, from calendar and system data.

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OPTIONS · TRADE-OFFS

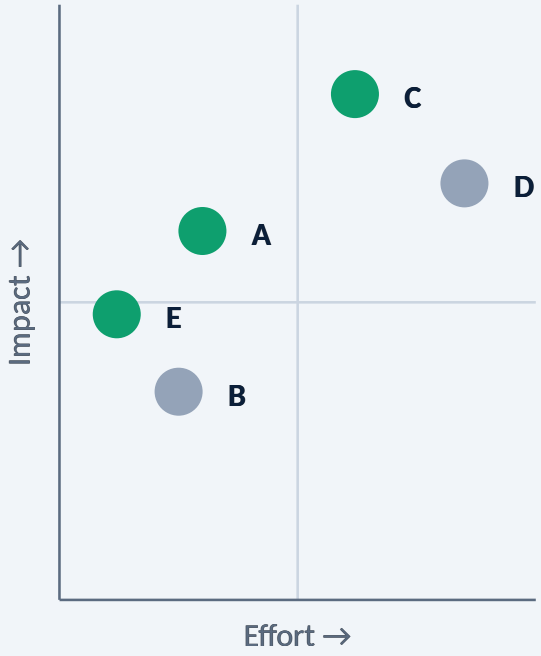
Three redesign levers beat a model upgrade at today's ≈2x; the upgrade still pays in support

Lever	What changes in the work	Forces fixed	Cost · speed	Main risk
A · Tier the checking Review by risk, not by default **RECOMMENDED**	Low-risk output sampled, high-risk fully reviewed; tests and match rules do the first pass	2	Low 1 quarter	Errors slip through low-risk tiers
B · Upgrade the model Newer model or agent tier	Faster drafting, coding and matching; more tasks within reach	1	Low–med Weeks	More output for the same reviewers to absorb
C · Redesign the flow Fewer hand-offs and approvals **RECOMMENDED**	Higher approval limits, review WIP limits and smaller PRs, close inputs due earlier	3	Medium 1–2 quarters	Pushback from other teams and controls owners
D · Give agents whole steps AI owns simple cases fully	AI resolves routine contacts and runs reconciliations; people take exceptions	1, 3 (part)	Med–high 2+ quarters	Quality slips unseen, as at Klarna in 2025
E · Plan the freed hours Decide where saved time goes **RECOMMENDED**	Name destinations before rollout: backlog, quality work, new services or hiring plans	4	Low Weeks	Hours leak into email and meetings

Forces (slide 6): 1 ceiling · 2 checking · 3 coordination · 4 moving target. Cost and speed are the author's estimates. Fund agents (D) once tiered checking holds quality.

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IMPACT VS. EFFORT



Teal = recommended. Positions are the author's estimates at today's ≈2x.

RECOMMENDATION · BUSINESS CASE

Fund whichever saves more hours: redesign saves ~1.7x a model upgrade's hours in software and finance

WHAT WE'D DO

1. **Map the hours first**

   Two weeks of time sampling per workflow (doing, checking, judgement, coordination) replace the (H) splits.

2. **Run the next-dollar test**

   Compare the hours a 2x better model saves with the hours a redesign saves. Fund the bigger; re-run at each model release.

3. **Redesign where the test says so**

   Tier the checking, cut hand-offs and approvals, and name where freed hours go: software and finance now.

BUILDS ON WHAT EXISTS

The inputs already exist: handle time and transfers in service, DORA lead-time metrics and Git logs in engineering, close calendars and APQC benchmarks in finance.

HOW I'D SIZE IT · SOFTWARE, PER 100 HRS

11.7	+	2.8	−	8.8	=	+5.7 hrs
⅓ of coordination (35)		40% of checking (7.0)		½ of AI-assisted work (17.5)		Above 0: fund redesign

Tiles 1–2: hours a redesign saves; tile 3: hours a 2x better model saves. Inputs I'd ask for: time-sampling splits, queue and approval waits, checking minutes per AI output.

THE TEST BY WORKFLOW · AUTHOR'S ESTIMATES

Workflow	AI-assisted · checking · coord. (H)	2x model saves	Redesign saves
Support	22.5 · 6.8 · 25	**11.2 hrs**	11.0 hrs
Software	17.5 · 7.0 · 35	8.8 hrs	**14.5 hrs**
Finance	20 · 10.0 · 40	10.0 hrs	**17.3 hrs**

1.7x Hours a redesign saves vs. a 2x better model, in software and finance; support is a toss-up.

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PILOT · HOW WE'D KNOW IT WORKED

Three 12-week pilots, with the same AI in both arms and three pre-set bars, decide where redesign scales



Metric	Good	Bad	Why this bar
End-to-end hours per unit (resolution, PR lead time, close days)	≥ 12% fewer than control	< 5% fewer	The model expects 13–19% fewer; 12% allows for slippage. Under 5% isn't worth the change effort.
Checking minutes per AI output	≥ 30% less	< 10% less	Risk tiers target 40% less. Under 10% means everything is still fully reviewed by default.
Freed hours put to a named use	≥ 60%	< 30%	85% of AI users already re-spend saved time (Denmark), but email time doubled (ActivTrak): naming the use decides the value.

Guardrail: CSAT and repeat contacts, escaped defects and incidents, post-close adjustments no worse than control. Bars are proposals to calibrate against the internal baseline in weeks 1–2.

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HAND-OFF CRITERIA

Scale when all four hold:

1. SOP written, one owner per redesigned step
2. KPIs live on the monthly operating review
3. Four straight weeks (two closes) at the "good" bar
4. Risk tiers signed off by QA or controls

RE-RUN THE TEST

At every model release, upgrade only where a 2x better model saves more hours than redesign. Support is first in line (11.2 vs. 11.0 hrs).

RISKS

The call flips if AI removes checking and coordination, not just doing

Risk	Why it matters	Mitigation / what we watch
Agents start removing steps	If AI routes work, approves low-risk items and chases inputs, coordination shrinks and upgrades pay again.	Re-run the test at each model release; judge agent pilots by steps removed, not tasks sped up.
Checking gets cheap	Models that verify their own work cut checking. Even halved, AI-alone ceilings reach only +46–71%.	Track the rework share (Workday: ~37% today); re-test if it falls below 10%.
Cutting checks too far	Errors cost more than the hours saved: Klarna rehired people; 26% of executives say AI errors reached boards or outsiders (Workiva).	Risk tiers, weekly audit samples and a quality guardrail in every pilot.
Missing the moving target	More ambitious work doesn't show up as more units per hour, so gains can look smaller than they are.	Track where freed hours go and what that work is worth.
The splits are estimates (H)	Support's call flips with a 10-point shift; software and finance hold under any 10-point shift.	Two weeks of time sampling before any budget moves.

WHAT WOULD CHANGE OUR MIND

We'd change course if:

1. **Output tracks models**
   Team throughput rises with each model generation, without redesign.
2. **Checking stops scaling**
   Rework drops under 10% of time saved, or auditors accept sampling.
3. **Agents own outcomes**
   AI resolves most contacts end to end at CSAT parity.

APPENDIX · MEASURED VS. ASSUMED

The software and finance calls survive any 10-point error in the time splits; support's does not

Input	Value used	Basis	Type
Today's speed-up on AI's share	2x	Measured range: 1.27x (Google RCT), 2.26x (lab RCT), 5.5x (Klarna chats)	(H) within (M)
Support split: AI · judgement · coordination	45 · 30 · 25	Admin, meetings and routine tasks ≈ 2/3 of agents' day (Salesforce); 19% of calls transferred (SQM)	(H) anchored
Software split	35 · 30 · 35	Coding 16% of developer time (IDC); meetings ≈12%, PR review ≈5% (Microsoft study, 484 devs)	(M) coding; rest (H)
Finance split	40 · 20 · 40	56% blame waiting on other teams (Ledge, n=100); median 6 days to close, 10 to reports (APQC)	(H) anchored
Checking time per unit of AI-assisted work	15 · 20 · 25%	Set so checking eats 30% / 40% / 50% of time saved; average 40% ≈ Workday's "nearly 40%"	(H) calibrated to (M)
Redesign: coordination removed	One third	APQC median to top quartile: close 25% faster, management reports 40% faster	(H) anchored
Redesign: checking removed	40%	Low-risk output sampled instead of fully reviewed	(H)
"Flat" line	Next doubling < 5%	Author's threshold for diminishing returns	(H)

Model: hours per unit = AI-assisted work ÷ speed-up + checking + judgement + coordination (Amdahl's law applied to a workflow). Splits are % of no-AI hours.

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NOTES

Robustness

Software and finance calls hold under any 10-point shift in the splits, any starting speed-up of 1.5x or more, and checking 50% higher or lower. Software ties if redesign removes only a fifth of coordination.

Left out

Queue effects: if review time rises 91% (Faros), software's ceiling falls from +39% to +31%. AI taking over judgement tasks. The value of more ambitious work.

SOURCES

Sources

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NOTE
Model and estimates are the author's own. Vendor studies are directional.

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